

Raghav Chhabra

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EDUCATION

Thapar Institute of Engineering and Technology, Patiala

Aug 2023 – May 2027

Bachelor of Engineering in Computer Engineering; CGPA: 9.0/10.0

ACHIEVEMENTS

- **Finalist** at #TECHMCODE by Tech Mahindra; among the **top 82 finalists** from 30,000+ participants nationwide (top 0.27%).
- Secured **AIR 2654** in Naukri Campus Young Turks 2025 (**Coding, Data Science & AI track**); scored **98.78 percentile** in Round 1 among 20,000+ participants.
- **National Semi-Finalist** in Flipkart GRID 7.0, selected from over 160,000 participants nationwide.
- Selected among the **Top 450** from 20,000+ applicants for the AlgoUniversity Technology Fellowship; selected for the **McKinsey Forward Program** (McKinsey & Company).
- Ranked among the **top 15,000** of 5M+ users on LeetCode; solved **1000+ DSA problems**, demonstrating strong mathematical and algorithmic aptitude.

WORK EXPERIENCE

Data Engineering Intern — Axis Bank, Business Intelligence Unit

Mumbai | May 2026 – July 2026

- **Owned** the design and development of a scalable **PySpark/Spark SQL** pipeline for preparing, validating, and reconciling large datasets to compute customer-level **financial and credit-risk metrics** across bureau request (**730M records**) and response (**10B records**) tables.
- Designed a customer-level summary layer capturing historical business-month data via custom statistical logic, eliminating redundant recomputation and supporting risk reporting cycles.
- **Cut pipeline execution time 90% (1 hr → 4–5 min)** by identifying process bottlenecks and re-architecting the job to cache intermediate datasets in Spark session memory.
- Built **approved automation tools** utilizing a JSON-driven dynamic configuration framework to remove hardcoded logic; automated formatted Excel report generation, streamlining communications for business stakeholders.

AI/ML Intern — OutriX

Remote | July 2025 – Aug 2025

- Designed and trained a custom CNN using PyTorch for CIFAR-10 image classification across 10 categories, building a full end-to-end pipeline from raw data ingestion to evaluated output.
- Automated dataset preprocessing and training workflows using optimized data loaders, mitigating manual steps and significantly improving overall pipeline control and efficiency.

PROJECTS

Time-Series Forecasting Engine using Machine Learning

GitHub | Demo

- Built an end-to-end **quantitative time-series forecasting pipeline** on millions of records; engineered 80+ predictive signals to support accurate forecasting and variance analysis.
- Benchmarked 7 regression models (Random Forest, XGBoost, etc.); achieved **99.53% forecasting accuracy** with rigorous out-of-sample validation.

Book Recommendation System

GitHub

- Engineered a personalized recommendation engine using TF-IDF Vectorization and SVD-based collaborative filtering, achieving an **RMSE of 0.84** on the Goodbooks-10k dataset.
- Modeled user-item interactions as bipartite graphs and applied the **A* search algorithm** to optimize recommendation paths and accuracy.

TECHNICAL SKILLS

Languages	Python, C, C++, SQL (MySQL)
Data Analysis	Process Improvement, EDA, Statistical Analysis, Data Visualization, Reporting, Pipeline Automation
Machine Learning Tools & Libraries	Regression, Classification, Clustering, Deep Learning (ANN, CNN), NLP Microsoft Office (Excel, PowerPoint, Word), PySpark, Pandas, NumPy, Scikit-learn, XGBoost, PyTorch
CS Fundamentals	Data Structures & Algorithms, OOP, DBMS, Operating Systems

RELEVANT COURSEWORK & INTERESTS

Coursework: Probability & Statistics | Data Structures and Algorithms | Artificial Intelligence | Machine Learning